

GATE

CRASH COURSE

CS/IT

Subject

Data Structure & Programming

Lecture No. 01

Abhishek Sir



Topics *to be covered*

- 1 Array 1-D
- 2 2-D array
- 3 3-D array
- 4





Array

- *Array is the Linear Data Structure.*
- *Arrays a kind of data structure that can store a **fixed-size, sequential** collection of elements of the same type (homogeneous).*

Array: Random Access ← Advantage
formula: $O(1)$
fixed size
Same data type



1-DArray

Elements are stored in consecutive memory location





Array Terminology

- 1 BA. Starting Address
- 2 Index of first element LB
- 3 Index of last element UB
4. Size



Array

Elements are stored in consecutive memory location

$A[1 \dots 10]$

$A[1]$	$A[2]$	$A[3]$	$A[4]$	$A[5]$	$A[6]$	$A[7]$	$A[8]$	$A[9]$	$A[10]$
1000	1002	1004	1006	1008	1010	1012	1014	1016	1018



Array

- **Base address:** Base address of the array is the *address of the memory where first element is stored*. **Base address of the array is 1000.**



Array

- *Lower bound:* Index of the first element in the array. In the above array 1 is lower bound. In C array lower bound is always 0.



Array

- *Upper Bound: Index of the last element* of the array, the above array is having upper bound as 10. In C language of array is declared as `int a[100]`, Upper bound is 99.



Array

- *Size of each element:* Size of each element is depends on the *data type*, the above array having size is 2 unit hence the first element is at 1000 second element is at 1002

int 4B

char: 1B

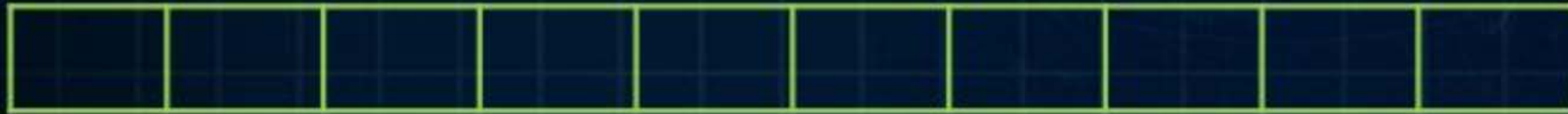
float 4B

Question

Consider the above array $A[1 \dots 100]$. Base address of the array is 1000 and each element occupies 2 Bytes of space. What is the address of $A[78]$.

$O(1)$

Random Access



$$BA + (i - LB) \times S_{\text{rec}} \quad \text{No. of element arranged}$$

$$1000 + (78 - 1) \times 2$$

$$1000 + 154 = 1154$$

Question

Consider the above array $A[-5 \dots 100]$. Base address of the array is 1000 and each element occupies 2-Bytes of space. What is the address of $A[78]$.

$$BA + \underbrace{(C - LB)}_{\substack{\uparrow \text{No of} \\ \text{elements arranged}}} \times S$$

$$BA + (C - LB) \times S$$

$$1000 + (78 - (-5)) \times 2$$

$$1000 + 83 \times 2 = \underline{1166}$$

C - program

LB - 0



Number of elements in this array?

Diagram illustrating the calculation of the number of elements in an array range $A[LB \dots UB]$ using the formula:

$$\boxed{UB_1 - LB_1 + 1}$$

Annotations:
 - LB (Lower Bound) points to -5 in $A[-5 \dots 10]$.
 - UB (Upper Bound) points to 10 in $A[-5 \dots 10]$.
 - UB_1 and LB_1 are the first elements of the range.
 - The formula is labeled "No. of Rows" and "No. of columns".

Examples:

- $A[-5 \dots 10] \rightarrow 10 - (-5) + 1 = 16$
- $A[10 \dots 19] \rightarrow 19 - 10 + 1 = 10$
- $A[0 \dots 16] \rightarrow 16 - 0 + 1 = 17$
- $A[2 \dots 19] \rightarrow 19 - 2 + 1 = 18$



Number of elements/rows/columns/frames Arranged?

$A[1 \dots 100]$

Address $A[78]$

$A[1] \dots A[77]$ - arranged

Before $A[78]$

$(78 - LB)$ elements arranged

$$78 - 1 = \textcircled{77}$$



Generalized formula:

- Given array $A[lb \dots ub]$
- Lower bound is lb
- Upper bound is ub
- Base address is BA
- Size of each element is S
- Address of $A[i] = BA + (i - lb) \times S$



2-D Array

- Consider the two dimensional array $A[1..4][1..4]$ Row, Column

	lb_2	Col. 1	2	3	4	ub_2
Row-1	lb_1	$A[1][1]$	$A[1][2]$	$A[1][3]$	$A[1][4]$	
2		$A[2][1]$	$A[2][2]$	$A[2][3]$	$A[2][4]$	
3		$A[3][1]$	$A[3][2]$	$A[3][3]$	$A[3][4]$	
ub_1 4		$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$	

$A[1..4][1..4]$
↑ Row ↑ Column



2-D Array

- Row Major Order ← C-program
- Column Major Order

A[1][1]	A[1][2]	A[1][3]	A[1][4]
A[2][1]	A[2][2]	A[2][3]	A[2][4]
A[3][1]	A[3][2]	A[3][3]	A[3][4]
A[4][1]	A[4][2]	A[4][3]	A[4][4]



Row Major Order

lb_1

lb_2

$A[1][1]$	$A[1][2]$	$A[1][3]$	$A[1][4]$
$A[2][1]$	$A[2][2]$	$A[2][3]$	$A[2][4]$
$A[3][1]$	$A[3][2]$	$A[3][3]$	$A[3][4]$
$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$

ub_2

Row major order

$A[4][3]$

$4^{th} (4-1) = 3$ Rows

Row -1				Row -2				Row -3				Row -4			
$A[1][1]$	$A[1][2]$	$A[1][3]$	$A[1][4]$	$A[2][1]$	$A[2][2]$	$A[2][3]$	$A[2][4]$	$A[3][1]$	$A[3][2]$	$A[3][3]$	$A[3][4]$	$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$
1000	1002	1004	1006	1008	1010	1012	1014	1016	1018	1020	1022	1024	1026	1028	1030



2-D Array

- In Column major order elements are arranged column wise. Addresses are given below each element.

OS + COA

	lb_2			ub_2
lb_1	A[1][1]	A[1][2]	A[1][3]	A[1][4]
.....	A[2][1]	A[2][2]	A[2][3]	A[2][4]
.....	A[3][1]	A[3][2]	A[3][3]	A[3][4]
ub_1	A[4][1]	A[4][2]	A[4][3]	A[4][4]

Column -1

Column -2

Column -3

Column -4

A[1][1]	A[2][1]	A[3][1]	A[4][1]	A[1][2]	A[2][2]	A[3][2]	A[4][2]	A[1][3]	A[2][3]	A[3][3]	A[4][3]	A[1][4]	A[2][4]	A[3][4]	A[4][4]
1000	1002	1004	1006	1008	1010	1012	1014	1016	1018	1020	1022	1024	1026	1028	1030



2-D Array

- In Column major order elements are arranged column wise. Addresses are given below each element.

lb_2 ub_2

lb_1 ub_1

$A[1][1]$	$A[1][2]$	$A[1][3]$	$A[1][4]$
$A[2][1]$	$A[2][2]$	$A[2][3]$	$A[2][4]$
$A[3][1]$	$A[3][2]$	$A[3][3]$	$A[3][4]$
$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$

$A[4][3]$ 4th Row, 3rd column

Column -1

Column -2

Column -3

Column -4

$A[1][1]$	$A[2][1]$	$A[3][1]$	$A[4][1]$	$A[1][2]$	$A[2][2]$	$A[3][2]$	$A[4][2]$	$A[1][3]$	$A[2][3]$	$A[3][3]$	$A[4][3]$	$A[1][4]$	$A[2][4]$	$A[3][4]$	$A[4][4]$
1000	1002	1004	1006	1008	1010	1012	1014	1016	1018	1020	1022	1024	1026	1028	1030

$$A[LB_1 \dots UB_1][LB_2 \dots UB_2]$$

Address $A[i][j]$

$$BA + [(j - LB_2) \times (UB_1 - LB_1 + 1) + (i - LB_1)] \times \text{size}$$

columns
completed

No. of elements
in column

No. of element
in last column

Question

Consider the following array $A[1 \dots 100][1 \dots 100]$. Base address is 1000 and each element occupies 2-Bytes of space. What is the *Multiple Row* address of $A[50][49]$ in row major order?

$$BA + \left[\underbrace{(50-1)}_{\text{\# of Rows}} \underbrace{(100-1+1)}_{\substack{\uparrow \\ \text{\# element in} \\ \text{each Row}}} + \underbrace{(49-1)}_{\substack{\uparrow \\ \text{\# No. of elements} \\ \text{arranged in} \\ \text{50}^{\text{th}} \text{ Row}}} \right] \times \text{Size}$$

$$1000 + [49 \times 100 + 48] \times 2$$

$$1000 + 4948 \times 2$$

$$1000 + 9896$$

10896

Question

Consider the following array $A[1...100]$
 $[1...100]$. Base address is 1000 and each element
occupies 2-Bytes of space. What is the address of
 $A[50][49]$ in column major order?

$$BA: 1000 \quad LB_1 = 1$$

$$Size = 2B \quad UB_1 = 100$$

$$i = 50 \quad LB_2 = 1$$

$$j = 49 \quad UB_2 = 100$$

$$BA + \left[(j - LB_2) \times (UB_1 - LB_1 + 1) + (i - LB_1) \right] \times Size$$

$$1000 + \left[(49 - 1) \times (100) + (50 - 1) \right] \times 2$$

$$1000 + 4849 \times 2 = 1000 + 9698 = 10698$$

$$A[LB_1..UB_1][LB_2..UB_2]$$

$$A[i][j] = BA + \left[\underset{\substack{\uparrow \\ \# \text{ of Rows crossed}}} {(i - LB_1) \times (UB_2 - LB_2 + 1)} + \underset{\substack{\uparrow \\ \text{No. of element in} \\ \text{each Row}}} {(j - LB_2)} \right] \times \text{Size}$$

$\uparrow$
 No of element
 in Last Row

Question

An array $[-2...2], [3...7]$ of elements and each element occupies 2 memory cells, the starting location is 1000. if the location of $A[0][5]$ using Row-Major-Order is X and using Column-Major-Order is Y then $X+Y$ is .

Column major order $A[0][5]$ $BA + [(j - LB_2) \times (UB_1 - LB_1 + 1) + (i - LB_1)] \times S$

$A[-2..2][3..7]$

$$LB_1 = -2$$

$$UB_1 = 2$$

$$LB_2 = 3$$

$$UB_2 = 7$$

$A[0][5]$

$$i = 0$$

$$j = 5$$

$$BA : 1000, S : 2$$

$$1000 + [(5 - 3) \times (2 - (-2) + 1) + (0 - (-2))] \times 2$$

$$1000 + [2 \times 5 + 2] \times 2$$

$$1000 + 12 \times 2 = 1024$$

$$X + Y = 2048$$

Question

An array $[-2...2], [3...7]$ of elements and each element occupies 2 memory cells, the starting location is 1000. if the location of $A[0][5]$ using Row-Major-Order is X and using Column-Major-Order is Y then $X+Y$ is .

Row major order: $BA + [(i - LB_1) \times (UB_2 - LB_2 + 1) + (j - LB_2)] \times size$

$A[-2..2][3..7]$

$$LB_1 = -2$$

$$UB_1 = 2$$

$$LB_2 = 3$$

$$UB_2 = 7$$

$A[0][5]$

$$i = 0 \checkmark$$

$$j = 5$$

$$BA : 1000, size : 2$$

$$1000 + [(0 - (-2)) \times (7 - 3 + 1) + (5 - 3)] \times 2$$

$$1000 + [2 \times 5 + 2] \times 2$$

$$\boxed{1024}$$



Summary

Type of Matrix	Ordering	Address of $A[i][j]$
2-D Array	Row major Order	$BA + [(i - lb_1) \times (ub_2 - lb_2 + 1) + (j - lb_2)] \times S$
2-D Array	Column Major Order	$BA + [(j - lb_2) \times (ub_1 - lb_1 + 1) + (i - lb_1)] \times S$

Question

If the array is declared as A [5...20][4...10] and base address is 1000 and size of the element is 4 Byte then what is the address of A[i][j] in row major order

A $BA + [28i + 148 + 4j]$

$$BA + [(i - LB_1)(UB_2 - LB_2 + 1) + (j - LB_2)] \times size$$

B $BA + [14i + 74 + 2j]$

$$BA + [(i - 5) \times (10 - 4 + 1) + (j - 4)] \times 4$$

C $BA + [28i - 156 + 4j]$

$$BA + [(i - 5) \times 7 + (j - 4)] \times 4$$

D $BA + [14i - 74 + 2j]$

$$39 \times 4$$

(C)

$$[7i - 35 + j - 4] \times 4$$

$$BA + [28i + 4j - 156]$$

Question

Consider the above array $A[-5 \dots 100]$.

Base address of the array is 1000 and each element occupies 2-Bytes of space. What is the address of

$A[70]$?

(A) 1150 ✓

(B) 1250

(C) 1200

(D) 1350

$$LB = -5$$

$$UB = 100$$

$$i = 70$$

$$BA + (i - LB) \times Size$$

$$1000 + (70 - (-5)) \times 2$$

$$1000 + 75 \times 2$$

$$1000 + 150 = 1150$$

Question

Consider the following array $A[\overset{117}{-23 \dots 93}] [\overset{39}{-7 \dots 31}]$. Base address is 1000 and each element occupies 2-Bytes of space. Which of the following is TRUE ?

- (A) ☒ The number of rows is 117 in Array A ✓
- (B) ☒ The Array $B[0..116] [2..40]$ have same number of elements as the array declared in question ✓
- (C) ☒ Address of $A[0][5]$ is 3538 row major order ✗
- (D) ☒ Address of $A[0][0]$ is 3538 row major order ✗

$$\begin{aligned} \text{No. of Rows} &= 93 - (-23) + 1 \\ &= 93 + 23 + 1 = 117 \end{aligned}$$

No. of element in 2-D array

$$B: \text{No. of Rows} = 116 - 0 + 1 = 117$$

$$\text{No. of column} = 40 - 2 + 1 = \underline{39}$$

less than (2818)

Question

Consider the following array $A[-23 \dots 93] [-7 \dots 31]$. Base address is 1000 and each element occupies 2-Bytes of space. Which of the following is TRUE ?

(C) Address of $A[0][5]$ is 3538 row major order

$$\begin{aligned} \text{RMO} &: 1000 + ((0 - (-23)) \times 39 + (5 - (-7))) \times 2 \\ &= 1000 + [23 \times 39 + 12] \times 2 = 2818 \end{aligned}$$



3-D Array

Three dimensional array is declared as $A[1...3][1...3][1...3]$. Three dimensional array consists of multiple two dimensional array called as *frame*. In each frame elements can be organized as *row major order* or *column major order*. Here is an example how a three dimensional array.



3-D Array

C-programming RMO

$A[i][j][k]$

$A[1...3][1...3][1...3]$

↑
frame

↑
Row

↑
column

i^{th} frame :- No. of frames completed
 j^{th} Row :- No. of Rows completed
 k^{th} column :- No. of columns completed

111	112	113
121	122	123
131	132	133

211	212	213
221	222	223
231	232	233

311	312	313
321	322	323
331	332	333

3rd frame

3rd Row

2nd column

Frames or 2-D Array



3-D Array

$A[1..3][1..3][1..3]$

$$A[3][3][2] = 1000 + \left[\underbrace{(3-1)}_{\# \text{ Rows}} \underbrace{(3-1+1)}_{\# \text{ columns}} \times (3-1+1) \right.$$

3rd frame

3rd Row ✓

2nd Column

$$+ \underbrace{(3-1)}_{\# \text{ of columns}} \times (3-1+1)$$

$$+ (2-1) \Big] \times \text{Size}$$



3-D Array

$A[LB_1..UB_1][LB_2..UB_2][LB_3..UB_3]$

$A[i][j][k]$

$$BA + \left[(i - LB_1) \times (UB_2 - LB_2 + 1) \times (UB_3 - LB_3 + 1) \right. \\ \left. + (j - LB_2) \times (UB_3 - LB_3 + 1) + \right. \\ \left. (k - LB_3) \right] \times \text{Size}$$



3-D Array

$A[1..100][1..100][1..100]$

Add

$A[50][49][50]$

$BA = 1000$

$Sne = 2B$

$$1000 + [(50-1) \times 100 \times 100 +$$

$$(49-1) \times 100 +$$

$$(50-1)] \times 2B$$

$$1000 + [4900000 + 48 + 49] \times 2$$

$$1000 + [494849] \times 2$$

$$= 1000 + 989698$$

$$\boxed{990698}$$



Summary



1. 1-D. $BA + (1 - LB) \times S$

2. 2-D. RMO: $BA + \left[(1 - LB_1) \times (UB_2 - LB_2 + 1) + (j - LB_2) \right] \times S$

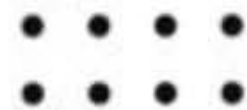
CMO + $BA + \left[(j - LB_2) (UB_1 - LB_1 + 1) + (1 - LB_1) \right] \times S_{re}$

3. 3-D. $BA + \left[(1 - LB_1) (UB_2 - LB_2 + 1) (UB_3 - LB_3 + 1) + (j - LB_2) \times (UB_3 - LB_3 + 1) + (k - LB_3) \right] \times S_{re}$

The word 'Thank' is written in a large, bold, yellow script font. A yellow arrow starts from the top of the 'T', extends horizontally to the right, and then curves downwards to point at the end of the word 'Thank'.

Thank

THANK



Keep Hustling!